Date

Expt. No. Page No.

NAME: Aastha Srivastava
COURSE: B.Tech CSE (FSD)
ROLL NO.: 2301350027

ASSIGNMENT - 4

SECTION - A

1) SRAM vs DRAM → SRAM uses flip flops to store data, making it faster, expensive and power consuming. It is used in cache memory.

DRAM uses capacitors to store data, needs refreshing, slower but cheaper. It is used as main memory.

2) Programs tend to access a small portion of memory
Type
- Temporable locality.
- Spatial locality.

3) Cache memory stores frequently used data/instructions closer to CPU. It reduces avg memory access time and improves performance.

4) Cause Indexing
- Direct mapped cache
- Set associative cache
- Fully associative cache

Teacher's Signature

Date

Expt. No. Page No.

5) Larger cache block size increases spatial locality (fewer misses) but increases miss penalty & may load unnecessary data

6) Associativity is the no. of locations where a memory block can be placed in cache. Higher associativity reduces conflict misses but increases hardware cost.

7) Larger cache size reduces miss rate & improves performance but increases cost, power & access latency.

8)

(i) Write-through: Data is written to both cache & main memory at the same time.

(ii) Write-Back: Data is written to cache first and main memory is updated only when the block is replaced.

9) AMAT = Hit time + (Miss rate × Miss penalty)

10) LRU block is replaced first. The block which has not been used for the largest time is removed.

Teacher's Signature

Date

Expt. No.

Page No.

11) memory is divided into multiple modules and accessed simultaneously in a round robin fashion. It increases memory bandwidth and improves performance.

12) Tracks are concentric circular paths on disk surface. Sectors are sub divisions of tracks used to store data blocks.

13) Flash memory is a non-volatile memory that retains data even when power is off. Used in SSDs, pendrives, memory cards. Faster than magnetic disks and has no moving parts.

14) I/O devices share the same address space as memory. Devices are accessed using normal memory instructions.

15) I/O devices have a separate address space. They are accessed using special I/O instruction.

16) CPU continuously checks the status of I/O device & controls the data transfer. CPU is busy waiting.

17) Device interrupt the CPU when it is ready. CPU can perform other tasks meanwhile. Better CPU utilization.

Teacher's Signature

Date

Expt. No. Page No.

SECTION - B

1)

Registers
↕
Cache
↕
Main Memory (RAM)
↕
Secondary storage (HDD / SSD)

Data flows b/w different levels upper levels are faster and smaller.
Lower levels are slower but larger in capacity.

2)

Address		
Tag (t bits)	Index (i bits)	offset (b bits)

cache lines

	Valid	Tag	Data Block (2^b bytes)
0			
1			
⋮			
$2^i - 1$			

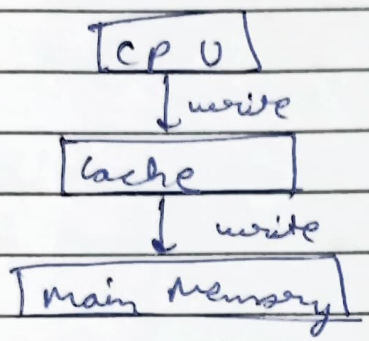
Each memory block maps to only one cache line using index.

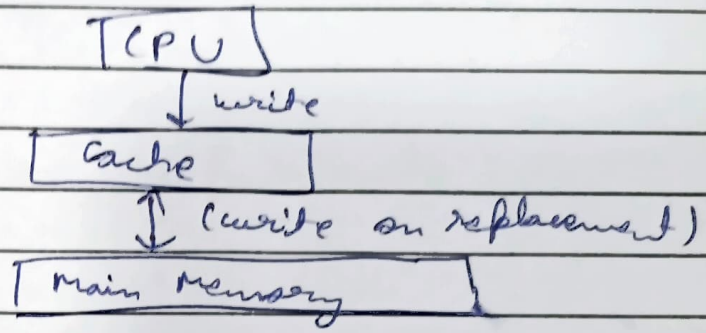
Teacher's Signature

Date

Expt. No. Page No.

3) (i) Write - through (ii) Write - Back

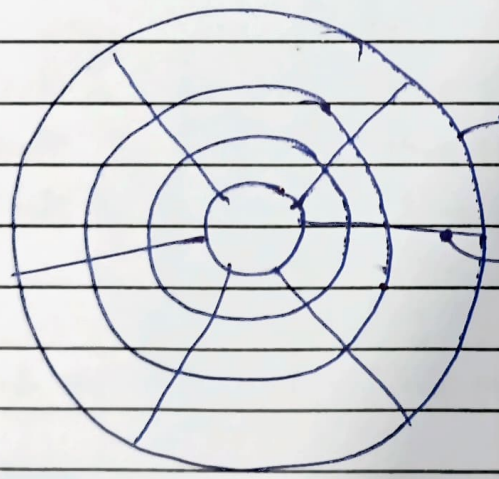




Each write updates both cache & main memory.

write updates only cache Main memory updated when block is replacement.

4)



→ Tracks (concentric circles)

→ Sectors (divisions of tracks)

SECTION - C

1) Temporal locality, because the same location is accessed again and again.

Teacher's Signature

Date

Expt. No. Page No.

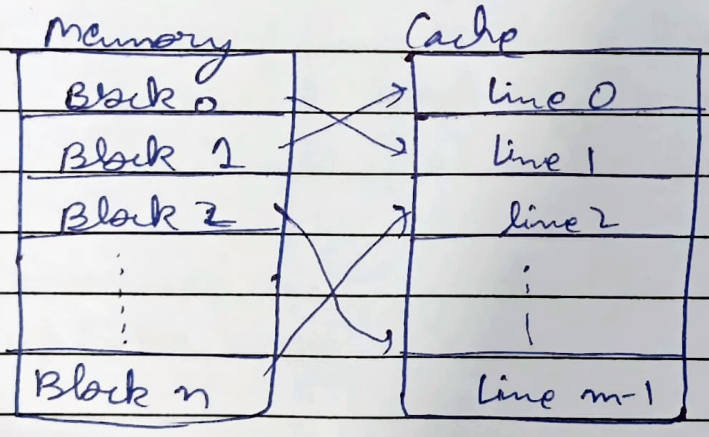
2) Larger blocks increase miss penalty (more data to be transferred) and may load unnecessary data (cache pollution).

3) On every write, data is written to both cache & main memory.

4) Direct Memory Access (DMA).

SECTION-D

1)



Mapping: ~~[illegible]~~

Mapping: memory block is placed in cache line using index. Tag is used in index correctness.

Teacher's Signature

Date

Expt. No.

Page No.

2) [AMAT = Hit time + (Miss rate × Miss penalty)].

Factors affecting performance
- lower hit time
- lower miss rate
- lower miss penalty.

These reduces AMAT and improve performance.

SECTION E

1) Two assumptions while explaining cache operation
- Fixed block size is assumed.
- Single level cache is considered.

2) Increases memory traffic because every write updates main memory.

3) Cache mapping is most difficult because it involves address break down (tag, index, offset) and mapping logic which is conceptual & needs practice.

Teacher's Signature